

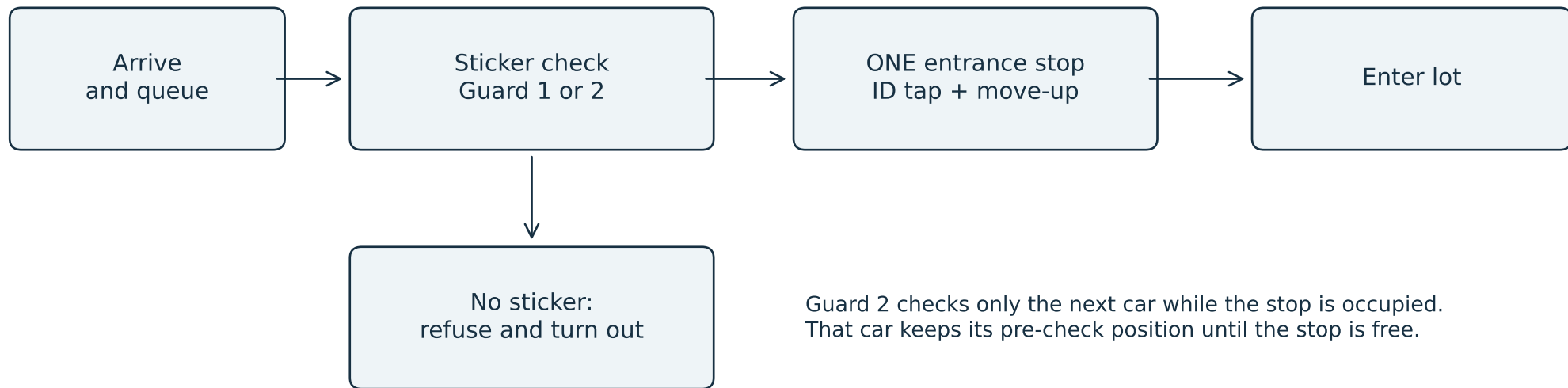
Should a second guard work during the morning?

Compare one guard all day, plus a second guard until 09:00, or plus a second guard until 10:00.

Test 20% fewer arrivals, normal demand, and 20% more arrivals. The aim is to measure the staffing benefit under stated assumptions.

The submitted proposal described independent booths and planned field measurements. This revision uses guards sharing one reader and estimated inputs.

Assumed entrance, not a confirmed campus layout



Inputs are estimates, not field measurements

Expected arrivals: about 402 cars/day, peaking at 100 cars/hour. Sticker check: 4 seconds; ID tap and barrier: 4 seconds; move-up: 3 seconds.

An estimated 5% of cars have no sticker. Refusal averages 45 seconds. The spillover threshold is 8 cars on the approach.

All values and their basis remain in the input CSVs. The arrival and activity distributions are assumptions; no campus fit was performed.

How we checked the program

27 recorded checks passed, including hand-calculated examples, car accounting, arrival order, closing rules, and one-reader limits.

Independent queue calculations checked each car's one-guard delay. Separate M/M/1 and M/G/1 benchmarks checked mean waiting under known conditions.

Nine combinations were run with 30 days each. Matching inputs make staffing comparisons fair. Empty demand, high demand, and a slow reader were also tested.

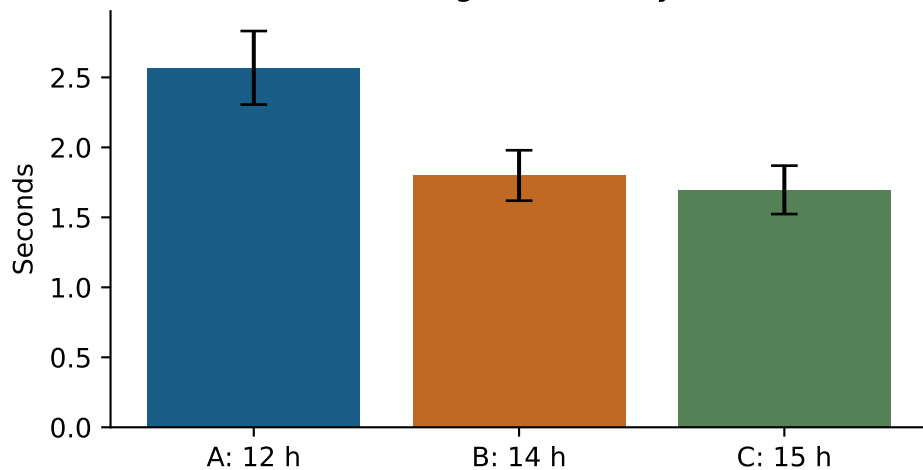
Schedule comparison at normal demand

Normal demand. 30 simulated days per schedule. Full confidence intervals are in the notebook and CSV exports.

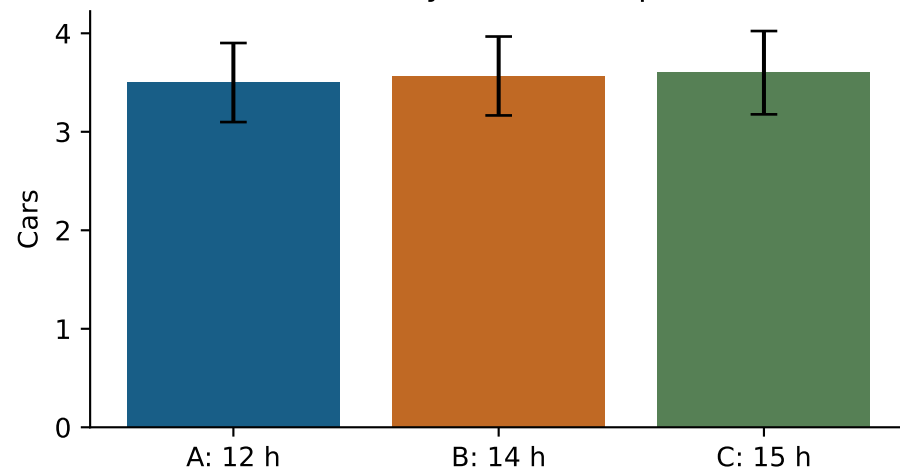
Schedule	Guard-hours	Delay (s)	Max queue	Guard use	Spillover (min)
A	12	2.57	3.50	5.79%	0.00
B	14	1.80	3.57	4.96%	0.00
C	15	1.70	3.60	4.63%	0.00

Estimated normal demand | 30 days per schedule | bars show mean and 95% CI

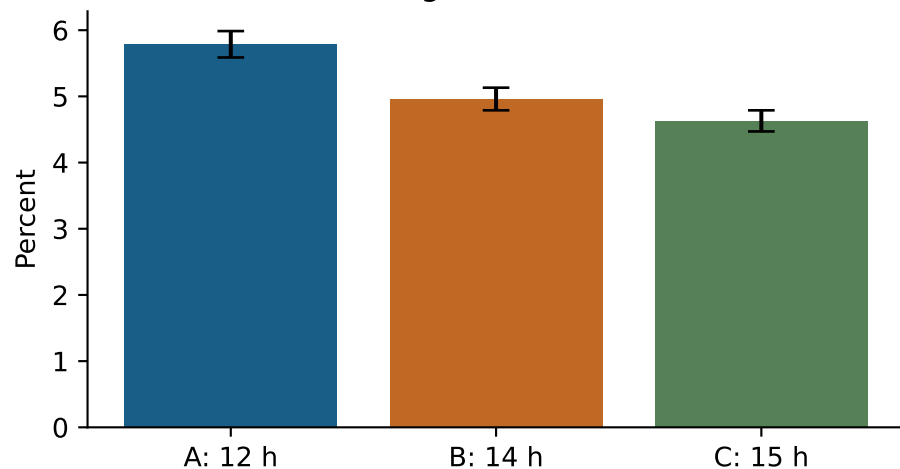
Average total delay



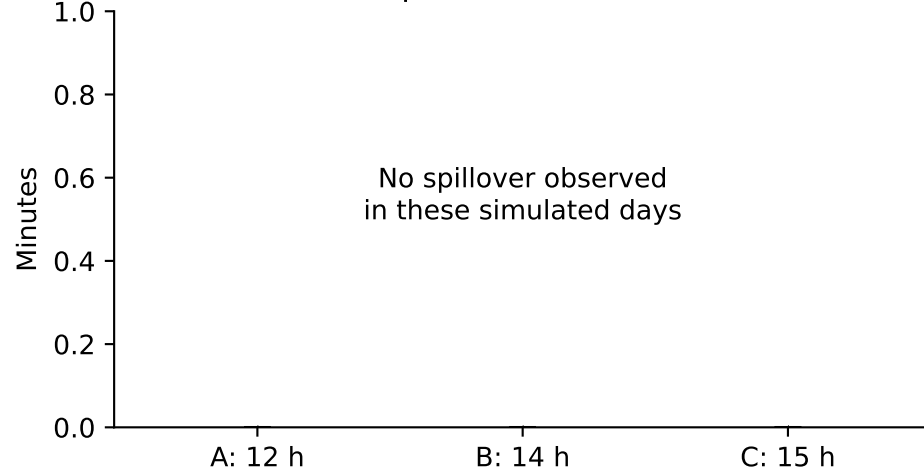
Mean daily maximum queue



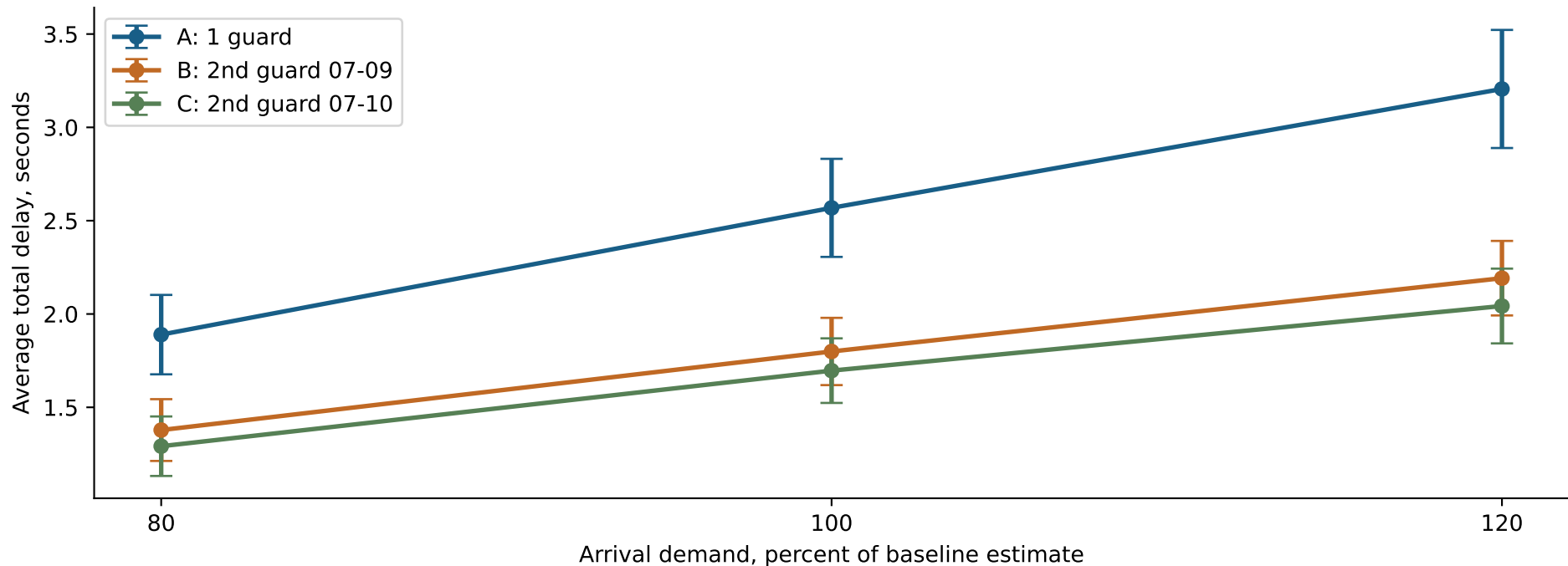
Active guard utilization



Spillover duration



Do the staffing comparisons change when arrivals change?



What the evidence supports

At normal estimated demand, A averaged 2.57 seconds of total delay. The next slides show the measured staffing benefits and confidence intervals.

The largest Wilson upper limit for spillover probability among study combinations is 11.4%. Zero observed events do not imply zero risk.

Confidence intervals describe variation between simulated days. They do not establish that estimated inputs match the campus.

Recommendation and limitations

Across 80%, 100%, and 120% demand, the largest mean delay was 3.21 seconds. No spillover was observed in the nine study combinations. This does not prove zero risk. Under these estimates, retain A as the starting staffing choice: the experiment has not established a need for more guards. Additional staffing has a measured time benefit, but its financial value was not modelled. This recommendation applies to the assumed arrangement and inputs. It is not a verified staffing recommendation for the real campus.

The layout, pre-check policy, refusal share, and approach capacity are not confirmed. No real queue measurements were available for validation.

Do not treat these results as proof of existing congestion, or as a financial case for hiring. Measure the inputs if access becomes available.